

my version control lecture and guides help me prepare a detailed and well explained note on this



Here are your lecture notes, expanded and structured so they are easy to study and follow. I've added a bit of extra context to the "why" and "how" so you understand exactly what the terminal commands are doing.

What is an SSH Key?

An **SSH (Secure Shell) key** acts as a highly secure, automated digital ID card. While usernames and passwords require you to log in manually, an SSH key allows your computer to securely communicate with remote servers (like GitHub) in the background.

Once set up, you can push and pull your version control repositories without needing to type your password every single time.

The Two Halves of a Key Pair

When you generate an SSH key, you are actually creating a **pair** of related files:

Public Key (`id_rsa.pub`): This is the padlock. You give this to outside services like GitHub. It is perfectly safe to share.

Private Key (`id_rsa`): This is the key that opens the padlock. It stays securely on your local computer. **Never share this file with anyone.**

Generating Your SSH Key

This process usually takes less than 20 minutes. The terminal commands below must be run exactly as shown, and the order matters.

① Open your terminal

Windows users must use Git Bash

Launch your command line interface. macOS and Linux users can use the default **Terminal**, but Windows users need to use **Git Bash** for these commands to work properly.

② Generate the key pair

Run the following command, making sure to replace the placeholder with the exact email address linked to your GitHub account:

Bash



```
ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
```

Note: -t rsa specifies the type of encryption, and -b 4096 ensures the key is highly secure by making it 4096 bits long.

③ Confirm the save location

Using the default is highly recommended

The terminal will prompt you for a file location to save the key. Simply press **Enter** to accept the default location.

This saves your keys in a hidden folder on your computer at `~/ .ssh/`.

④ Set a passphrase (Optional)

You will be asked to create a passphrase for an extra layer of security. If you want a completely automated experience without typing anything when you connect, just press **Enter twice** to leave it blank.

⑤ Verify your keys (Optional)

To confirm your keys were successfully created, navigate to the hidden `.ssh` directory and list its contents by running:

Bash



```
cd ~/.ssh  
ls
```

You should see `id_rsa` (your private key) and `id_rsa.pub` (your public key) listed.

⑥ Start the SSH agent

The `ssh-agent` is a background program that holds your private key in memory so it can be used for authentication. Start it up with this command:

Bash



```
eval "$(ssh-agent -s)"
```

You should see an output with a process ID (PID) number, confirming it is running.

⑦ Add your key to the agent

Finally, load your newly created private key into the agent so it can do its job:

Bash



```
ssh-add ~/.ssh/id_rsa
```